

[illegible]

		$[A \rightarrow a \cdot bAb, d]$ $[B \rightarrow a \cdot bc, c]$ $[C \rightarrow da \cdot da, a]$ $[D \rightarrow aa \cdot d, c]$ $[E \rightarrow ca \cdot c, a]$ $[F \rightarrow aAa \cdot, a]$ $[G \rightarrow ba \cdot, a]$ $[H \rightarrow Ha \cdot, c]$	
3	a	Write appropriate rules to construct a Syntax tree for the following set of statements: $P \rightarrow S$ $S \rightarrow \text{if} (C) S \mid \text{id} = \text{num}$ $C \rightarrow \text{id}$	5
	b	Write appropriate rules to generate 3-address code for the following statements: $S \rightarrow \text{id} = E$ $E \rightarrow E + T \mid T$ $T \rightarrow \text{id}$	5
	c	Append appropriate rules to the given CFG to count the number of occurrences of the substring ab in a string made up of a's and b's. $S \rightarrow Sa \mid Sb \mid a \mid b$ For example the string aabbaba contains two occurrence of ab, whereas the string baaaba contains only one occurrence of ab. Justify the working of your rules by carrying out the SDD over the input string : aabbaba	10 (5 + 5)
4	a	Locally optimize the following block by constructing a DAG for it: $a = b * c$ $d = b$ $e = d \times c$ $b = e$ $f = b + c$ $g = f + d$	5
	b	Construct a Control flow graph for the given intermediate code : (Note : Also introduce Entry and Exit nodes) $a := b$ L1: $b := c$ if (...) goto L4 $c := b$ L2: $d := a$ goto L1 L4: $b := a$ if (...) goto L2	5
	c	Provide quadruple representation for the set of intermediate code: a) $a = 5$ b) $b = a * 5$ c) Label L1 d) $x[t3] = 3$ e) goto L2	5

